

Kanu Madhok

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PROFESSIONAL EXPERIENCE

Walmart Data Ventures

Chicago, IL · Sep 2024 – Present

Senior Data Analyst

- Designed and shipped to production a self-service analytics system over 67 BigQuery tables with sub-agent orchestration and human-in-the-loop validation (Context Researcher → SQL Drafter → Validator → Devil's Advocate); 8 deterministic SQL rules and a 23-case golden-query evaluation suite testing correctness before release. Partnered directly with Product, Data Science, and business stakeholders to translate ambiguous questions into precise specs. Approved by Sr. Director and product leadership.
- Replaced ~20 distinct recruitment workflows I'd been running by hand 1–2× weekly with end-to-end automation on a 68-table BigQuery platform. Stratified-sampling pipelines recruited 280,000+ panelists across 29 categories with zero duplicates. A recurring 3-hour task now takes ~3 minutes of setup.
- Built a registry-governed semantic layer over BigQuery with strong type discipline: a single LLM call maps the question to strict JSON intent, a deterministic gate validates and canonicalizes it into a typed dataclass, and a deterministic compiler emits the SQL — same input, same output, every time, with the one probabilistic step isolated and regression-tested.
- Built and shipped an autonomous Jira ticket-resolution agent in modern, refactorable Python: 6-gate triage, MCP-driven context gathering, FAISS + BM25 retrieval over 11,000 historical SQL queries, self-healing execution with a 10-retry budget. Closed 400+ data requests across 8 categories over 15 months — cut routine turnaround from 30–60 minutes to under 10 minutes each.
- Built and deployed a headless KPI monitor on a 6-hour cron that detects threshold breaches, runs root-cause analysis, and posts Slack alerts; end-to-end observability via decision logging and BigQuery state persistence (no in-memory reliance). Designed a 3-tier event-driven v2 platform self-configuring via a BigQuery event registry — new event types ship without code deploys.
- Delivered 5 Power BI dashboards ranked top 1.4% of Walmart's 207,000 dashboards (1,064 views, 22 users); cut data-request turnaround from 1–24 hours to 5–10 minutes; Making a Difference Award.

University of Chicago — Data Science Institute

Chicago, IL · Jan – Sep 2024

Graduate Researcher, Data & Democracy Initiative

- Persona-driven multi-LLM (OpenAI + Gemini) chatbot for a 500+ participant donation experiment; controlled A/B tests persuaded 15% to donate to an opposing cause.

FTI Consulting — UChicago MSADS Capstone

Chicago, IL · Jan – Aug 2024

- RAG-to-proposal generator deployed on AWS EC2; cut proposal time ~30%; Best in Show — UChicago MSADS Capstone.

Innovare (EdTech Startup)

Chicago, IL · Oct 2022 – Mar 2023

Data Scientist

- Logistic-regression risk model + Looker dashboards for 10+ districts (5,000+ students); retention +7%, failures –15%.

SELECTED PROJECT

- Live deployed text-to-SQL copilot (Streamlit + Gemini 2.5 + LangChain-orchestrated RAG over historical queries, BigQuery execution, auto-visualization): live demo — <https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

SKILLS

Data engineering: BigQuery, dbt, SQL, Python, PySpark, Airflow, Databricks, MySQL; large-table pipelines, stratified-sampling, BigQuery scheduled queries, JSONL state stores

Engineering rigor: modern refactorable Python, typed-dataclass validation, deterministic compilers, golden-set / human-in-the-loop evaluation suites, CI/CD (pytest, version control), decision logging / observability

Agent stack: MCP server design, sub-agent orchestration, context engineering, RAG (FAISS + BM25 + RRF), headless cron agents, event-driven architectures

LLMs: Claude (Opus / Sonnet / Haiku via Wibey), Gemini (2.5 Pro / Flash via Vertex AI), OpenAI

APIs / triggers / glue: FastAPI, Cloud Run, Slack webhooks, cron, Playwright SSO

Viz: Power BI (DAX, XSLT), Tableau, LookML, Streamlit, Matplotlib, React Recharts

EDUCATION

The University of Chicago — M.S., Applied Data Science (GPA 4.0)

Chicago, IL

Loyola University Chicago — B.S., Information Systems & Economics (GPA 3.5)

Chicago, IL